S3-1

Perch v2: A foundation model for bioacoustics: Supervision is superb

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We present Perch v2, our latest bioacoustics foundation model, as well as lessons learned along the way. Chief among them is the bitter(n) lesson on machine learning for bioacoustics: supervised training is an extremely hard baseline to beat when coupled with large amounts of in-domain, well-curated, and high-quality annotated audio recordings. Perch v2 prvides both off-the-shelf classification scores for thousands of vocalizing species as well as strong embeddings for search and transfer learning. In this new release, we expand to the multi-taxa setting and obtain state-of-the-art performance on the BirdSet and BEANS benchmarks. Perch also outperforms SurfPerch on the ReefSet transfer learning tasks, despite having no coral reef training data. In addition, we provide broad support for the bittern lesson, both from the theoretical literature and experimental evidence.

S3-2

Robots, acoustics and AI, oh my: Integrating sensing and soundscapes analyses to quantify coral reef health and biodiversity hotpots

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Coral reefs harbor some of the highest biodiversity on Earth. Their rich soundscapes are biologically vital and can provide a means of tracking community health. Yet these reefs are facing immense climate stressors and are declining rapidly. Passive acoustics can provide a powerful, scalable tool for assessing change and reef health, but providing stakeholder-relevant information including detecting and classifying biological sounds, and thus assessing biodiversity and biological hotspots, remains a key challenge within this complex environment. We present results aiming to detect and classify coral reef fish calls, and then implement those analyses into actionably-relevant rapid assessments including a novel undersea robot (CUREE-Curious Underwater Robot for Ecosystem Exploration) built to detect and identify biological hotspots. Leveraging 60,000 labeled fish calls from a long-term (10-yr) acoustic and ecosystem study of U.S. Virgin Islands coral reefs, we trained a robust AI-based detector for reef fish sounds. The CNN detects and classifies over 100 Caribbean fish calls, providing insight into soundscape changes driven by climate-related stressors, and helps differentiate between reefs of varying health state, capturing variation in crepuscular activity and predicting coral cover, fish abundance, and species richness. The detector was then implemented in the CUREE robot for autonomous discovery of reefs and non-intrusive audio-visual, spatial quantification of fish densities and biological hotpots. Together, these methods offer new means to assess reef health spatially and temporally. Implemented into a novel rapid acoustic assessment, they provide a key step toward providing actionable information to stakeholders monitoring for habitat change and weighing resource management.

S3-3

Passive acoustic gliders are effective monitoring tools to support dynamic management of whale-vessel strikes

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Dynamic management is intended to mitigate human impacts on wildlife when and where they overlap with at-risk species. We studied the performance of mobile underwater passive acoustic gliders as near real-time monitoring assets for dynamic management to reduce the risk of whale-vessel strikes. Critically endangered North Atlantic right whales (NARWs) were acoustically detected in near real-time on 45 days out of 1,035 glider survey days in the Gulf of St. Lawrence, which triggered 285 days of mandatory 10-knot vessel speed limits in three Canadian Dynamic Shipping Zones (DSZs). We found a high degree of interannual and seasonal variation in NARW acoustic occurrence and vessel slowdowns in the DSZs. Gliders provided continuous monitoring and were more effective than aerial surveillance at triggering slowdowns by a factor of approximately 2 and 5 during the fall and summer, respectively. They were less effective in the spring, when whales did not call as frequently. Our results provide unambiguous empirical evidence that gliders are effective monitoring platforms that can enhance dynamic ocean management goals globally.

S3-4

Advancing passive acoustic monitoring for the management of commercially and ecologically important fish species

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Passive acoustic monitoring (PAM) is an increasingly valuable tool for studying and managing fish populations, particularly given the rise in anthropogenic pressures and climate-driven changes in marine ecosystems. Over 1,000 fish species are currently known to be soniferous, with new species identified regularly. However, the application of PAM remains constrained by limited information on species-specific calls in situ and the analytical burden of processing large acoustic datasets. This presentation synthesizes recent advancements in instrumentation and analytical techniques designed to overcome these challenges. First, the development and deployment of portable audio-video array platforms for in situ localization and identification of calling individuals are presented. These systems have recently been used off the west coast of Canada to detect and characterize previously undocumented sounds from five rockfish species, including species of conservation concern. These data provide new opportunities to enhance monitoring within Rockfish Conservation Areas (RCAs), where traditional survey methods are logistically constrained. Second, the integration of machine learning algorithms into PAM workflows is discussed. Convolutional neural networks have been developed and applied to detect species-specific calls in extensive passive acoustic datasets, enabling fine-scale analysis of temporal and spatial patterns. Case studies include the monitoring of Melanogrammus aeglefinus (haddock) in the Gulf of Maine, revealing patterns of seasonal occurrence and reproductive activity. Finally, recent work targeting Clupea harengus (Atlantic herring) in the northeastern United States is highlighted, with emphasis on evaluating PAM's utility for tracking spawning activity and informing fisheries management.

S3-5

Acoustic biogeography of European marine forests

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Marine forests (e.g., kelp, seagrass meadows) are among the most productive and biodiverse habitats, providing essential ecosystem services. Yet face increasing pressures from human activities and climate change, they are also particularly vulnerable. These forests shelter numerous species whose sounds rhythmically mark daily and seasonal cycles. Despite the potential of ecoacoustics in acquiring information on cryptic species of dense habitats, the biophonic component of temperate marine forests remains poorly documented, with only 3% of ecoacoustic studies carried out in these ecosystems. This study describes the biogeography of fish acoustic communities of three temperate marine forests (kelp, Zostera and Posidonia meadows) at the European scale (Atlantic and Mediterranean coasts). During the European TREC expedition, continuous sound recordings were conducted over 48-72h at 17 sites. Using a standardized sound classification method, we provide the first detailed description of bioacoustic communities in European marine forests. We analyzed acoustic communities (composition, abundance, richness, biodiversity indices) and revealed habitat, intra- and inter-site variability. Acoustic communities were distinct between habitats, with greater site variability within seagrass meadows, compared to kelp, suggesting differences in seagrass connectivity. Site characteristics and sampling periods were considered, and relationships between acoustic communities and diel variation were examined. Our findings offer an unprecedented large-scale view of acoustic communities in underexplored habitats. This work reveals that acoustic biodiversity can contribute to habitat biogeography and infer information on the hidden animal assemblages, their dynamics and responses to environmental changes.

S3-6

Soundscape as an indicator of ecological characteristics in a temperate primeval forest

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Passive Acoustic Monitoring (PAM), combined with automatic species recognition and soundscape complexity analysis, has been proposed as an effective approach for providing reliable indicators of ecological characteristics, such as habitat type, condition, and species diversity. However, recent studies have often reported inconsistent results. In this study, we focused on birds as the primary source of biological sounds in the temperate, primary Białowieża Forest (Poland). We employed human-based point counts, soundscape recordings, automatic species recognition, and acoustic indices to explore the relationships between the soundscape and ecological characteristics. Our analysis covered two datasets: simultaneous soundscape recordings and bird counts by observers at 20 points, as well as year-round soundscape recordings at 40 points (30,000 hours of soundscape recordings). We found that traditional human-based surveys underestimated bird species richness compared to estimates obtained through automatic recognition with BirdNet. The soundscape complexity varied between forest types, despite no significant differences in bird species richness. The relationships between acoustic indices and bird species richness or abundance were highly dependent on the settings used for index calculations. In all the analyzed models, soundscape complexity was better explained by total bird abundance than by bird species richness. The analysis of daily and year-round changes in soundscape characteristics and bird species richness revealed habitat-specific patterns. Our study demonstrated that (PAM), when combined with automatic species recognition and soundscape complexity analysis, may effectively reflects ecological characteristics of primary temperate forests. However, using automatic methods without prior verification of their effectiveness may lead to misleading conclusions.